

Soft Margin SVM – Mathematics

- ✓ In practice, real-world data is often noisy, overlapping, or not linearly separable.
- ✓ That's where **Soft Margin SVM** comes in — it allows some **misclassifications** but still tries to maximize the margin.

1 Recall Hard Margin SVM

We had the optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w^T x_i + b) \geq 1 \quad \forall i$$

This assumes **perfect separation**.

2 Introducing Slack Variables (ξ)

To handle misclassifications, we introduce **slack variables** $\xi_i \geq 0$.

Now the constraint becomes:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

- If $\xi_i = 0 \rightarrow$ point is correctly classified and outside margin.
- If $0 < \xi_i < 1 \rightarrow$ point is inside margin but still correctly classified.
- If $\xi_i > 1 \rightarrow$ point is misclassified.

3 New Optimization Problem

We now balance two goals:

- Keep the margin large ($\|w\|^2$ small).
- Keep misclassification penalty (ξ_i) small.

The optimization problem becomes:

$$\min_{w,b,\xi} \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

4 Role of C (Regularization Parameter)

- CC controls the trade-off between **margin width** and **classification errors**.

➤ Large CC:

- Penalizes misclassification heavily
- Forces the SVM to correctly classify most training data
- May lead to **overfitting**

➤ Small CC:

- Allows more misclassifications
- Margin becomes wider
- More tolerant, helps **generalization**

5 Visualization Intuition

- Hard margin: perfectly straight boundary, **no mistakes allowed**.
- Soft margin: boundary is more flexible, **allows some errors**.

6 Final Form (Soft Margin SVM)

$$\min_{w,b,\xi} \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

➤ Quick Recap

- Soft margin SVM introduces **slack variables** ξ_i to handle misclassification.
 - Parameter C decides the trade-off between **maximizing margin** and **minimizing classification error**.
 - Hard Margin is just a **special case** of Soft Margin when $C \rightarrow \infty$ and $\xi_i = 0$.
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